

WATERTOWN, SD, USA

WMO: 726546

Lat: 44.905N

Lon: 97.149W

Elev: 533

StdP: 95.09

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14946

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.3	(c) -23.6	(d) -29.3	(e) 0.3	(f) -25.9	(g) -27.0	(h) 0.3	(i) -23.4	(j) 14.4	(k) -11.3	(l) 12.7	(m) -9.3	(n) 3.6	(o) 320	(p) 0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	31.9	22.8	30.2	21.7	28.7	20.8	24.4	29.7	23.3	28.2	22.2	26.9	6.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.7	27.5	21.7	17.4	26.5	20.5	16.2	25.3	76.7	29.8	72.1	28.2	67.6	26.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.5	(b) 11.2	(c) 10.0	(e) -30.5	(f) 35.1	(g) 3.4	(h) 1.7	(i) -32.9	(j) 36.3	(k) -34.9	(l) 37.4	(m) -36.9	(n) 38.3	(o) -39.4	(p) 39.6		
			DB													
			WB													
				-30.5	26.0	3.3	1.1	-32.9	26.8	-34.8	27.5	-36.6	28.1	-39.0	28.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-10.5	-8.7	-1.7	6.3	13.2	19.0	22.0	20.5	15.9	7.9	-0.4	-7.7
	DBStd	12.77	7.34	7.50	7.29	5.79	4.94	3.81	3.39	3.22	4.71	5.29	6.31	6.69
	HDD10.0	2664	637	523	366	138	25	1	0	0	7	103	316	549
	HDD18.3	4660	895	757	621	363	171	36	8	14	101	324	563	807
	CDD10.0	1351	0	0	4	26	125	271	372	326	184	39	3	0
	CDD18.3	304	0	0	0	1	12	57	122	83	28	2	0	0
	CDH23.3	2846	0	0	1	18	146	534	1161	699	262	26	0	0
	CDH26.7	873	0	0	0	3	36	152	405	203	69	4	0	0
(14) Wind	WSAvg	4.7	4.8	4.9	5.1	5.4	5.1	4.4	4.0	3.9	4.5	5.0	4.9	4.9
(15) Precipitation	PrecAvg	581	14	14	30	56	78	101	78	70	55	44	22	13
	PrecMax	1017	54	49	100	165	246	289	291	163	147	206	79	61
	PrecMin	275	0	0	3	2	10	16	13	13	6	1	0	0
	PrecStd	145	12	12	22	35	53	60	51	40	35	42	19	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	10.9	20.4	25.7	30.2	32.9	34.9	33.1	31.4	26.9	18.7	8.7
		MCWB	2.6	6.9	11.9	14.9	18.8	22.4	24.0	23.4	21.0	16.3	11.0	5.2
	2%	DB	2.8	6.7	15.3	21.8	27.4	30.2	32.4	30.8	28.8	22.3	14.6	4.9
		MCWB	0.9	3.8	10.2	12.6	18.0	21.1	23.6	22.6	20.1	14.6	9.0	2.0
	5%	DB	1.2	3.5	11.8	18.8	24.7	28.3	30.6	28.9	26.6	19.0	11.0	2.7
		MCWB	0.0	1.7	8.0	10.7	16.3	20.1	22.5	21.2	18.9	12.9	6.9	0.8
	10%	DB	-0.5	1.5	8.3	15.9	22.1	26.5	28.9	27.2	24.1	16.4	8.1	0.9
		MCWB	-1.4	0.2	5.0	9.4	14.7	19.0	21.6	20.3	17.8	11.7	4.7	-0.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	7.8	13.8	15.8	20.9	24.3	26.5	25.4	23.1	18.5	12.7	5.3
		MCDB	5.8	10.4	17.0	23.6	27.5	30.4	31.3	30.7	27.8	24.1	17.3	8.1
	2%	WB	1.1	3.9	10.7	13.4	19.0	22.8	25.0	23.8	21.6	15.8	9.7	2.5
		MCDB	2.5	6.2	14.5	20.1	25.3	28.2	30.2	28.8	26.5	20.2	13.7	4.4
	5%	WB	0.0	1.8	8.0	11.5	17.5	21.4	23.8	22.6	20.5	13.9	7.1	0.9
		MCDB	1.2	3.4	11.9	17.1	23.2	26.8	28.8	27.0	25.0	18.4	10.9	2.5
	10%	WB	-1.4	0.2	5.1	10.0	15.8	20.1	22.7	21.5	19.0	11.9	5.0	-0.3
		MCDB	-0.5	1.5	8.3	15.2	20.7	25.1	27.5	25.7	22.9	15.8	8.0	0.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.9	9.8	11.7	11.5	11.0	11.5	11.4	11.9	10.8	9.7	8.5
		MCDBR	10.7	11.1	14.6	16.7	15.1	13.3	13.0	13.0	14.2	15.4	14.4	11.0
	5% WB	MCWBR	9.2	8.7	9.5	9.2	8.0	6.6	6.0	6.4	7.0	8.9	9.3	8.6
		MCDWR	10.4	10.4	14.3	14.6	13.0	11.5	11.6	11.7	12.1	13.5	14.0	10.4
(40) Clear-Sky Solar Irradiance	taub		0.262	0.282	0.306	0.352	0.369	0.379	0.384	0.390	0.350	0.314	0.283	0.261
		taud	2.429	2.396	2.429	2.333	2.367	2.387	2.370	2.337	2.430	2.504	2.497	2.448
	Ebn at Noon		871	912	933	909	899	889	879	860	871	863	838	834
		Edh at Noon	70	88	99	119	119	117	118	117	98	79	66	63
(44) All-Sky Solar Radiation	RadAvg		1.81	2.88	3.79	4.83	5.58	6.25	6.66	5.53	4.31	2.80	1.86	1.35
	RadStd		0.19	0.29	0.37	0.42	0.41	0.41	0.24	0.46	0.31	0.35	0.18	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page